

ELEC-H550: Embedded Systems Security

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Session 2: Practicalities

Short Bio:

- Professor at Université libre de Bruxelles, Embedded Systems Design and Security
- Previously Research Manager at KU Leuven, DistriNet Group
- PhD on software verification from Uni. of York, UK
- Topics
 - Hardware & Software Co-Design for Security
 - Embedded Systems Security, Safety-Critical Systems
 - Secure Processors & Trusted Computing
 - Automated Software Testing and Formal Verification
 - Security and Privacy for Vulnerable People
 - Sustainability Aspects in ICT and Infrastructures



The Responsible Security and Privacy Lab @ ULB



Thesis Topics

You have to develop your own idea; these topics from the last years:

- Age-Appropriate Design of Privacy Policies
- Security and Privacy in Smart-Grid Energy Communities
- V2X Revocation and Enrolment
- Sustainability Assessment of Security Architectures
- Security Assessments in Embedded Control Systems
- Privacy and Security in IoT Things for Children
- Data Practices in Migration Aid

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Anything on...

- Resilient and privacy-preserving communications (with Iness Ben Guirat)
- Security and privacy for vulnerable people (with Apolline Zehner)
- RISC-V security extensions & formal verification (with Navid Ladner)
- Secure real-time systems (with Oscar Van Slijpe)
- Side-channels, ML/LLM leakage, fuzzing (with Abia Smahi & Cedrick Austa)
- sustainable security and privacy (with Jason Kayembe)

Internships & Seminars

Companies to reach out to:

- ... will be announced.

Seminars:

- PrivacyCamp (2026-10-13): <https://privacycamp.eu/>

The Plan for ELEC-H550

- 1 The information security landscape and the the role of safety, security, and data protection in embedded systems
- 2 Low-level vulnerabilities and defences in software and hardware
- 3 Vulnerabilities and defences in light-weight embedded systems
- 4 Automated detection, exploitation, and prevention of vulnerabilities in software
- 5 System security and secure hardware
- 6 Sustainability aspects in security and privacy engineering
- 7 Security assessment techniques
- 8 **Exercises:** Threat modeling, software fuzzing, exploitation techniques
- 9 **Project:** Your own research

Practicalities

- 24h Lectures (= 2 ECTS) + 60h Project & Exercises (= 2.5 ECTS)
+ 6h Seminar (= 0.5 ECTS)

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- **Course Materials and Announcements:**
uv.ulb.ac.be/ || cloud.timeedit.net/be_ulb/web/
- **Lectures:** Tuesdays 10:15 till 11:55 (Solbosch)
- **Exercises, Project, Seminars:** Tuesday 16:00 till 17:30 (Solbosch)
- **Grading:** Presentation & Questions (8 points) + Final Report (12 points)
- **Textbooks:**
 - *Security Engineering: A Guide to Building Dependable Distributed Systems*, 3rd Edition, Ross Anderson, 2020.
 - *Threat Modeling: Designing for Security*, Adam Shostack, 2014.

Practicalities: Project & Project Topics

Group work is encouraged, four milestones:

- Early October: Topic and research questions
- Early November: First results, literature list, etc.
- December: Presentations
- January: Reports

Practicalities: Project & Project Topics

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Topic Ideas:

- (Privacy) Thread Modeling of a mobile app or IoT device
- Security profile of a home IoT device
- Security and privacy in your own home-automation project
- Reproduce a known vulnerability or research paper
- Literature study on emerging trends in critical infrastructure or IoT, and the related security and privacy impacts
 - “Hackers Could Use IoT Botnets to Manipulate Energy Markets”
 - “Critical Bugs Expose Hundreds of Thousands of Medical Devices and ATMs”
- Software Fuzzing on embedded processors
- Comparison of security extensions in embedded processors

What makes a good project?

- Clear problem statement / research questions / objectives
- Methodology and information sources are explained
- The methodology is followed and individual steps or results are explained
- Conclusions address research questions or explain what the next steps towards answering the questions would be
- I can follow the argumentation that leads to the conclusion
- Technical depth, accuracy and completeness of the content
- Individual contribution of the student, creativity, problem solving, technical contribution
- Structure: abstract, introduction, elaboration, conclusions, references
- Citations: clearly identify what is your contribution and what comes from the sources
- Estimated workload: 40h per student

Practicalities: Evaluation

1	ELEC-H550 Report Grading	
2		
3	Points (max. 12) & Expectation:	k
4	1: Clear problem statement / research questions / objectives	
5	2: Methodology is justified and information sources are explained	
6	1: The methodology is followed and individual steps or results are explained	
7	1: Conclusions (or clearly indicated part elsewhere) address research questions or explain what	
8	1: You can follow the argumentation that leads to the conclusion	
9	4: Technical depth, accuracy and completeness of the content	
10	2: Individual contribution of the student, creativity, problem solving, technical contribution	
11	Bonus points for something remarkable?	
12	If not present:	
13	-1: Structure: abstract, introduction, elaboration, conclusions, references	
14	-1: Citations: clearly identify what is your contribution and what comes from the sources	
15	-1: Estimated workload: 40h per student	
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